MultiCLIP: Four-View Contrastive Learning for Metabolite Identification from Tandem Mass Spectra

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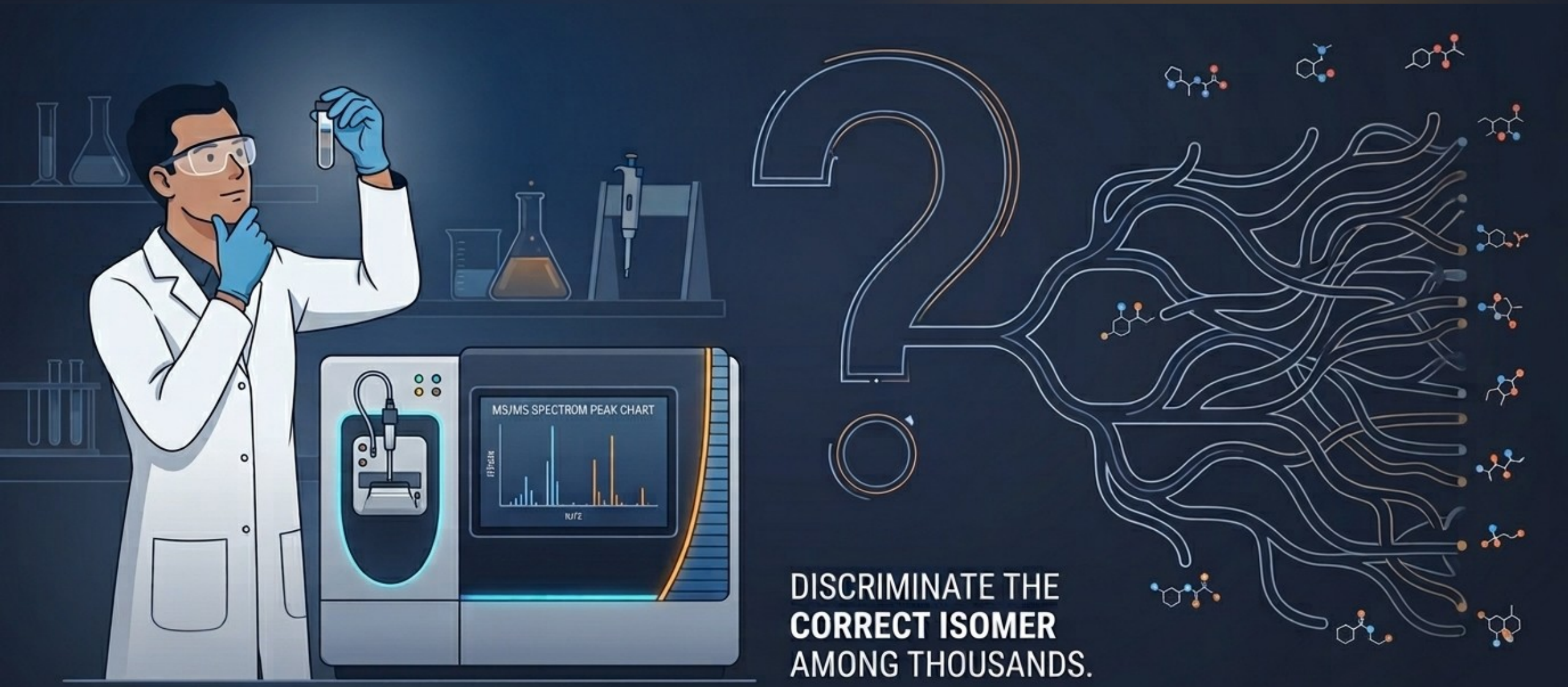
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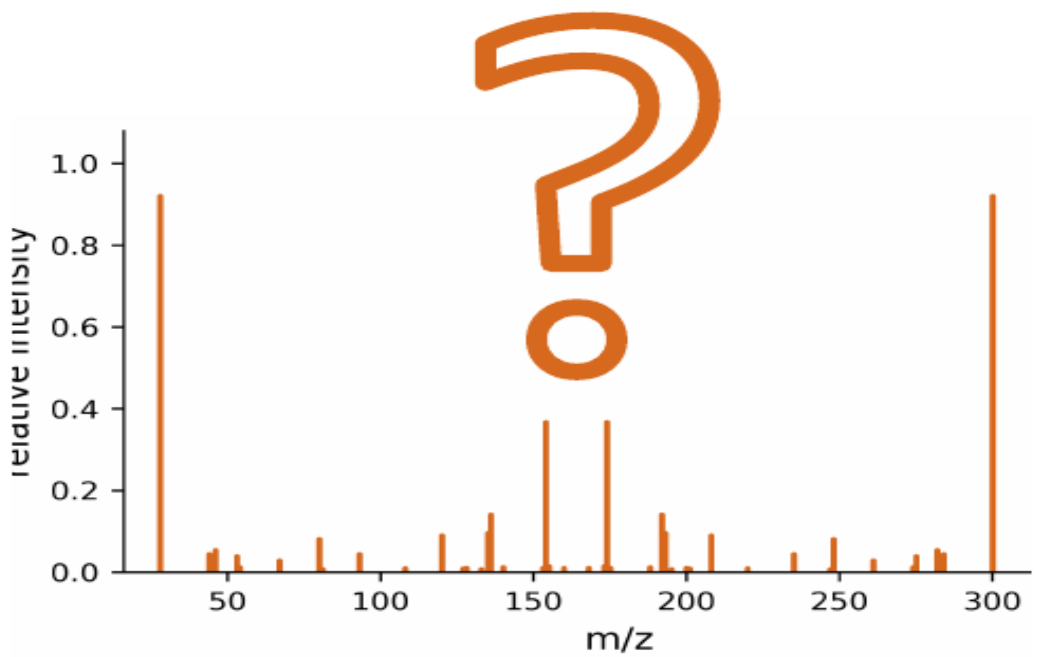
Contact: manuel.ruiz@urv.cat

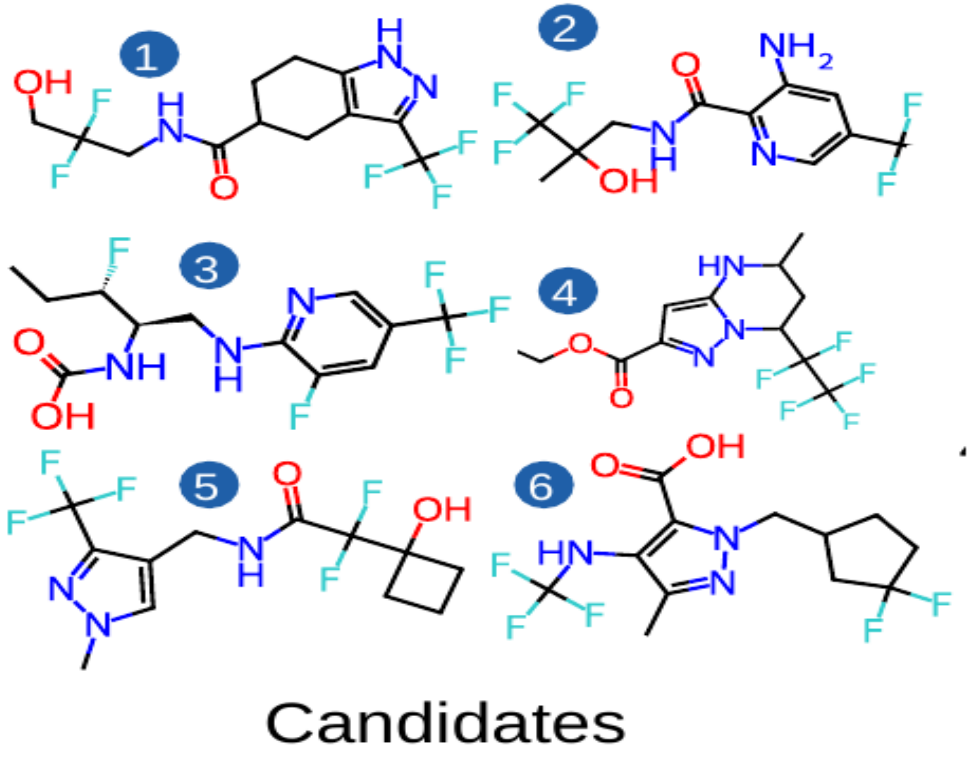


DISCRIMINATE THE
CORRECT ISOMER
AMONG THOUSANDS.

This spectrum was produced by one of these molecules.

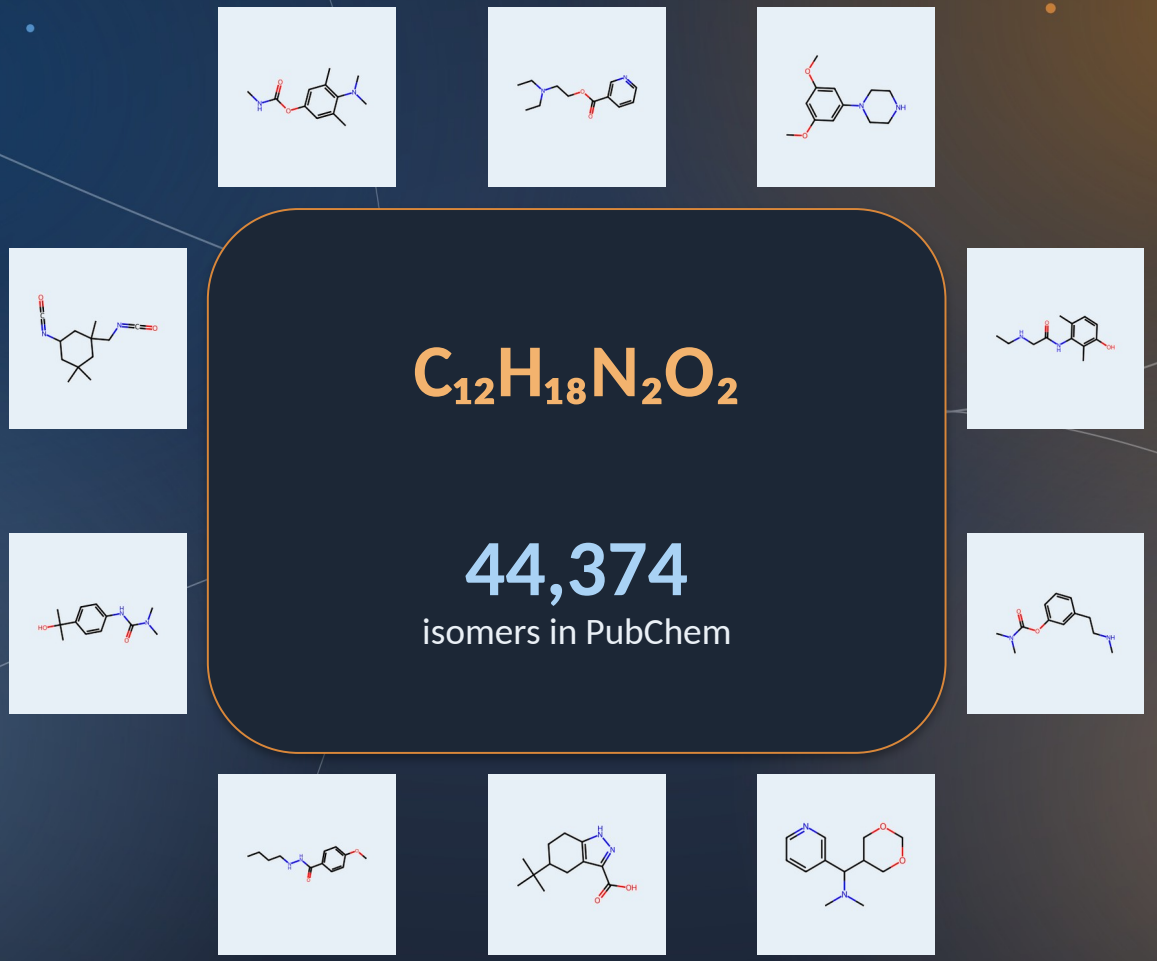
They share the same precursor mass and the molecular formula.





But how hard can get this problem?

You can write $C_{12}H_{18}N_2O_2$ from the precursor mass. You still do not know which compound is in the sample.



But how hard can get this problem?

And looking it up in a library barely helps.

2.3%

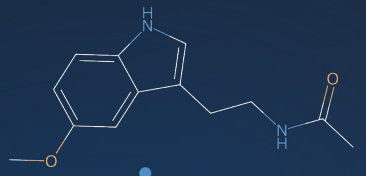
of experimental MS/MS spectra
are named by a GNPS library search



1 in 42 spectra gets a name. Martin et al., 2024

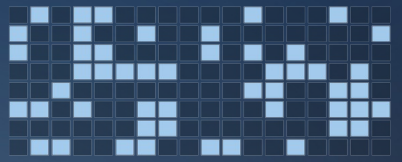
Aligning four views of a metabolite helps to identify the isomer from several candidates.







One metabolite · four views



2048-bit Morgan



Aligning four views of a metabolite helps to identify the isomer from several candidates.

01 INPUT

Tokenise every m/z peak directly, instead of current approaches

02 ENGINE

Align and fuse four views: graph view molecule, fingerprints, merged spectra, and single spectra.

03 HORIZON

Understand the latent geometry of the four views, to understand how to reconstruct new molecules.

How do you encode a spectrum?

01 **INPUT**

Tokenise m/z directly

02 **ENGINE**

Align and fuse four views

03 **HORIZON**

Understand the latent geometry

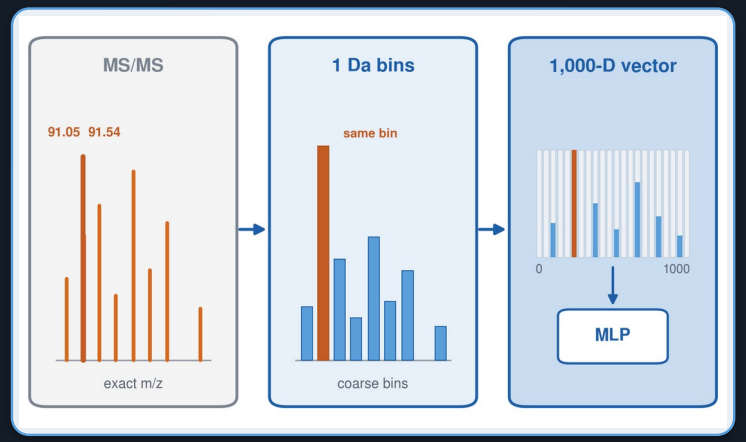
Current methods encode spectra in three different ways

The chemically richest tokens assume a precursor formula. The formula-free ones pay another price.

1 Da histogram

JESTR

Formula-free

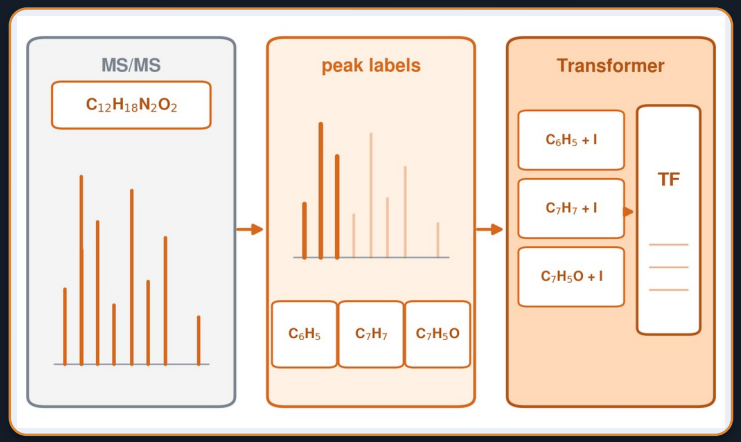


Exact mass is collapsed into 1,000 bins.
Isomer resolution is what you lose.

Subformula peak tokens

MIST · MVP · FLARE

Needs precursor formula

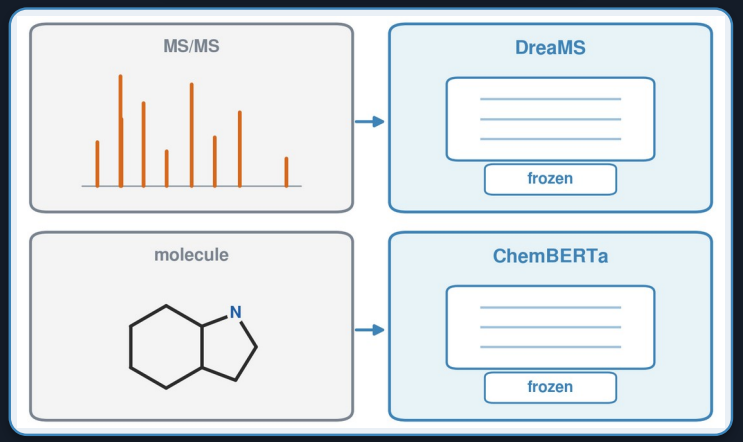


Each peak is labeled from the precursor formula.
A wrong formula poisons every token.

Foundation embeddings

MSAlign

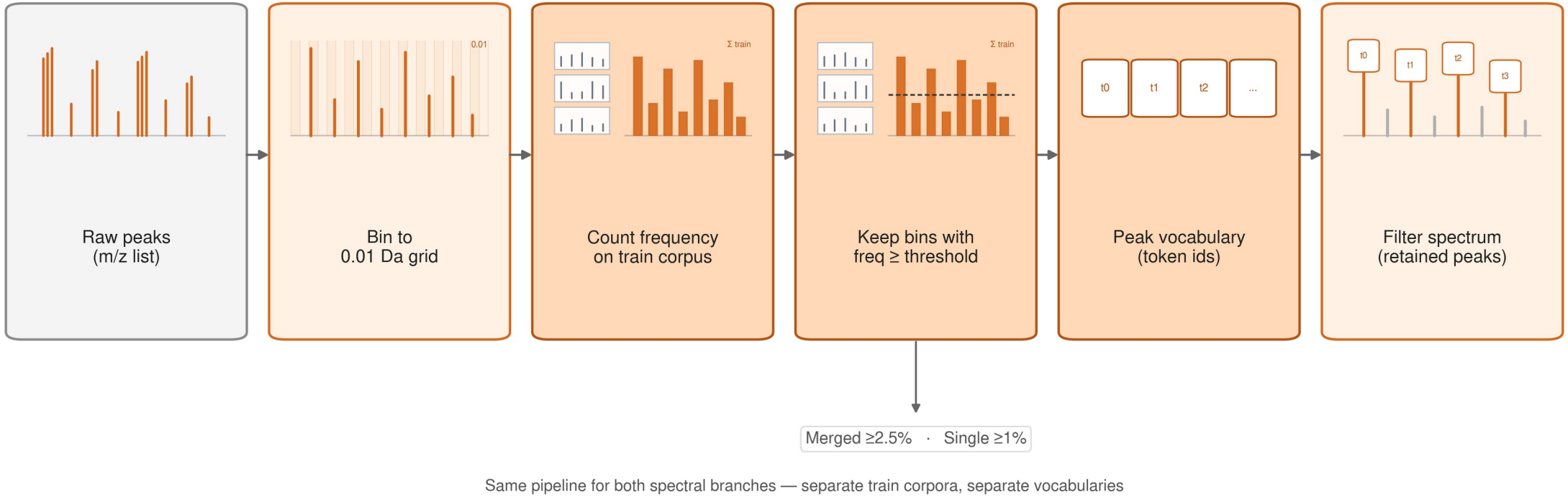
Formula-free



Frozen DreaMS + ChemBERTa, then MLPs.
No peak chemistry. No formula.

Build the vocabulary from recurrent and informative m/z peaks

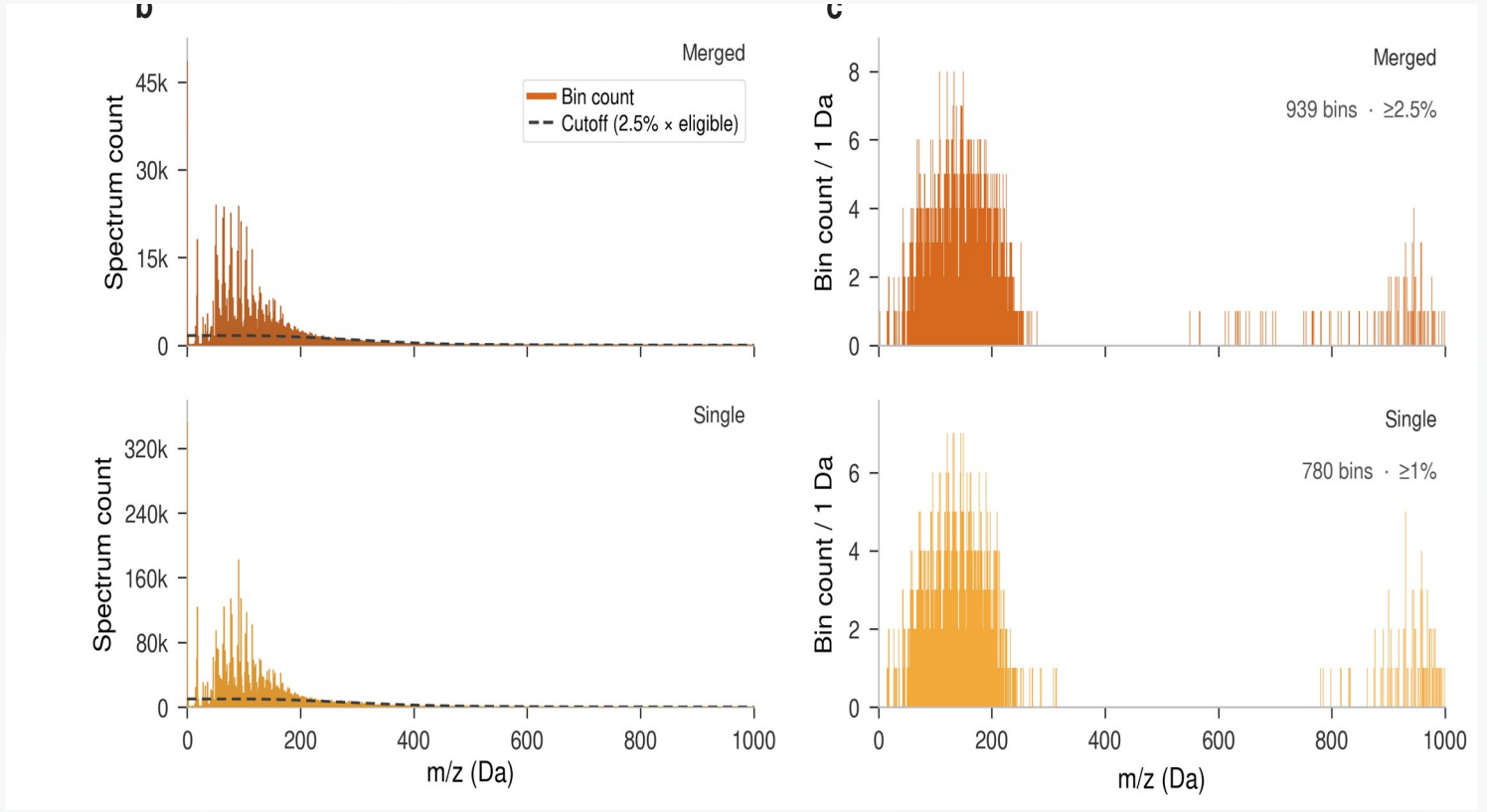
Frequency on the training corpus decides which peaks become tokens.



No precursor formula. Training corpus only.

The vocabulary concentrates on recurrent low-mass fragments

Paper evidence: occupancy thresholds select compact, data-driven token sets.



Merged

939

tokens · ≥ 2.5%

Single

780

tokens · ≥ 1.0%

Coverage

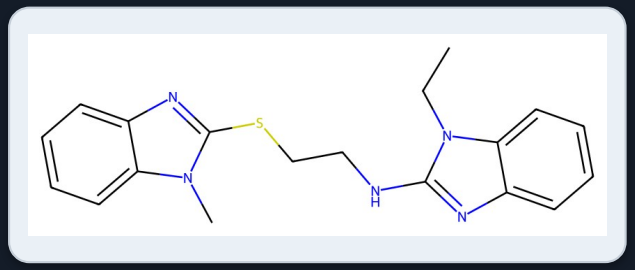
33%

of peak occurrences

No precursor-dependent subformulae.

Vocabulary filtering on one molecule

In-vocabulary peaks stay as tokens — rare peaks, and the precursor, drop out.



$C_{19}H_{21}N_5S$

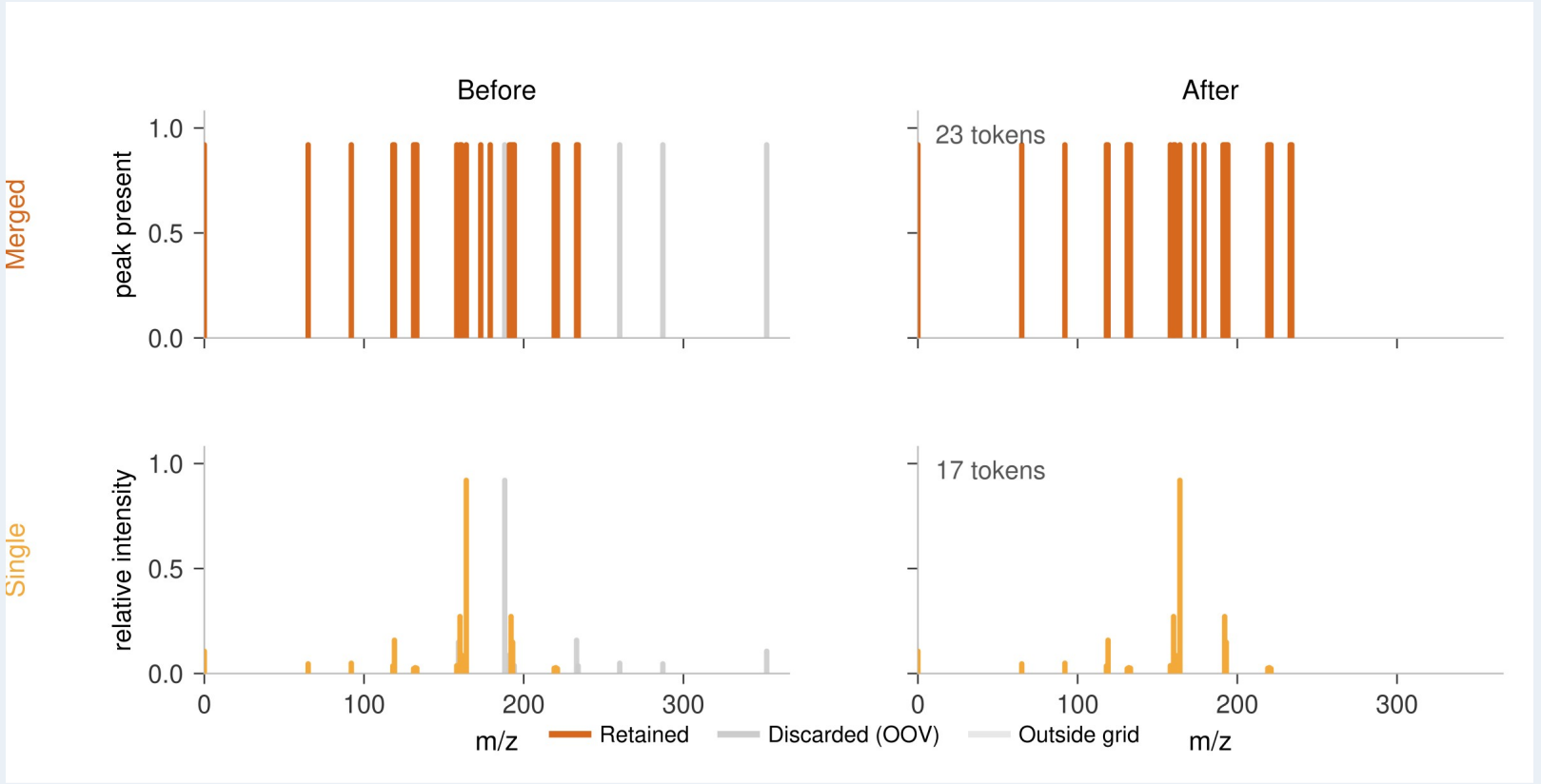
Merged

23 tokens

Single

17 tokens

Grey peaks = out of vocabulary



The grey peak near m/z 352 is the precursor ion — discarded as out of vocabulary.

How do four views of a molecule get aligned?

01 INPUT

Tokenise m/z directly

02 ENGINE

Align and fuse four views

03 HORIZON

Understand the latent geometry

The molecule and the experiment each have two complementary views

Connectivity + substructure. Stability + measurement-specific detail.

MOLECULE

2D graph

atom connectivity

GNN

MOLECULE

Morgan fingerprint

substructure pattern

MLP

512-D

shared
embedding space

SPECTRUM

Merged consensus

stable signature

Transformer

SPECTRUM

Single spectrum

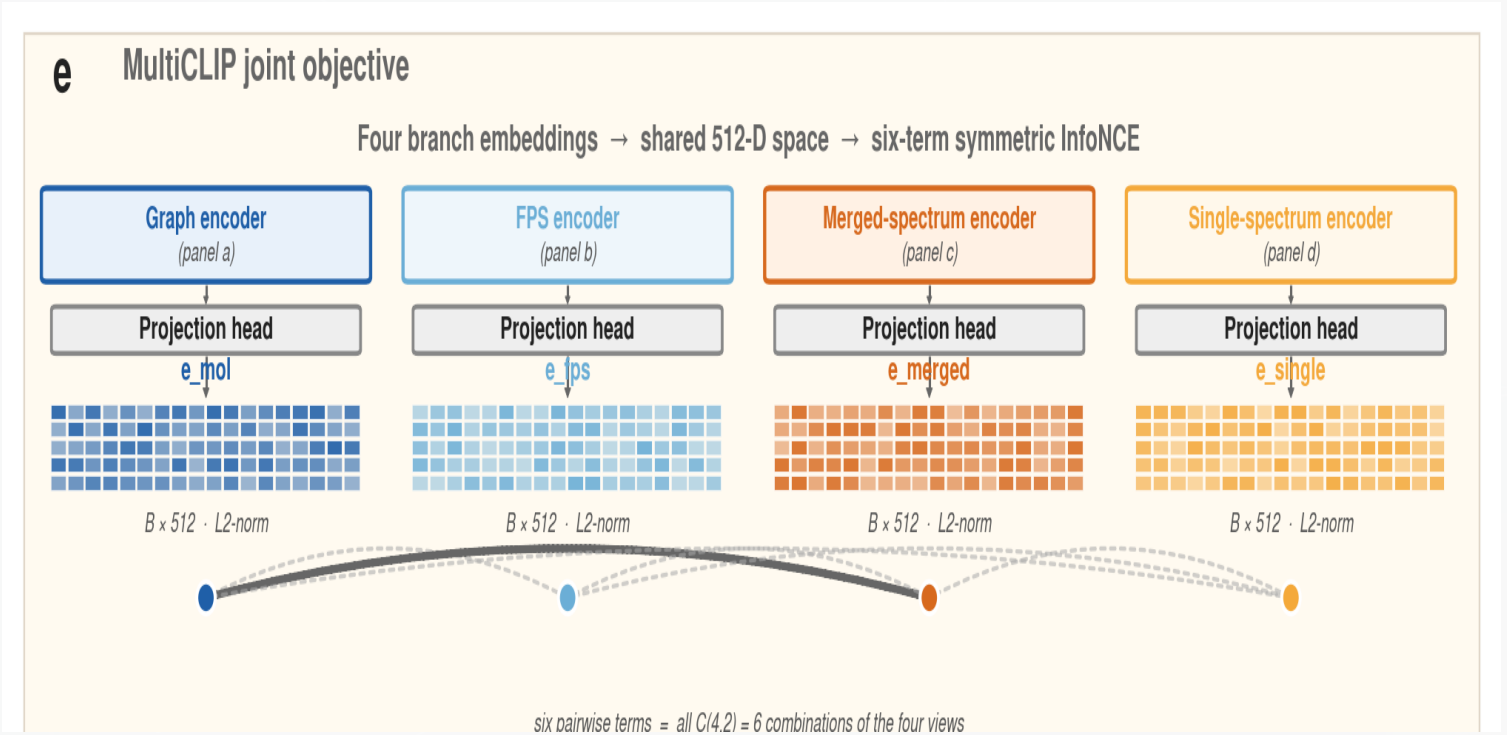
intensity + losses

Transformer

Same compound → four compatible embeddings

Every view is aligned with every other view

Four branches create six positive pairings per compound.



6 terms

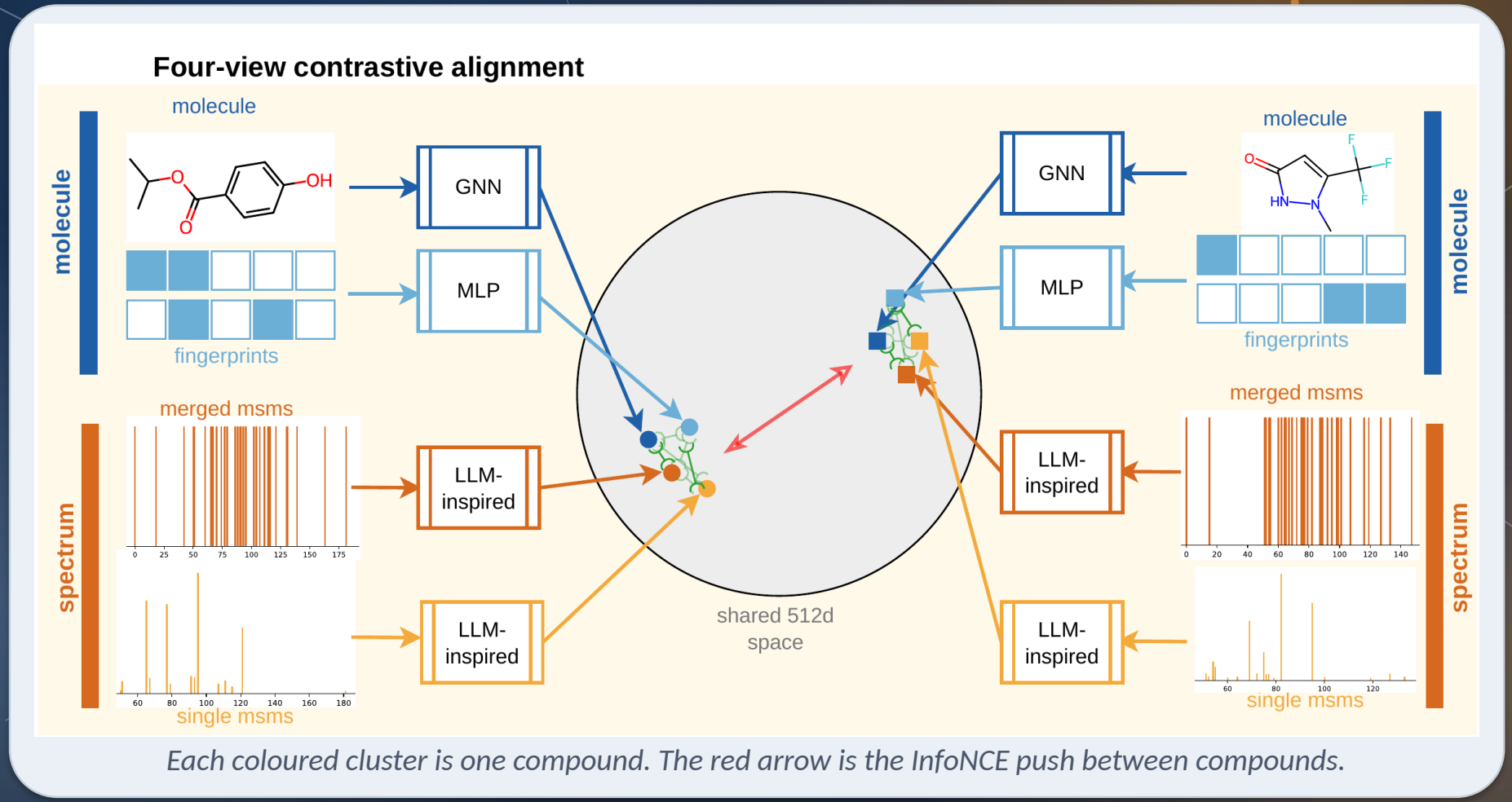
Each pair pulls
the same compound together
and pushes others apart.

Why this matters

No branch can specialise
in evidence invisible
to the other three.

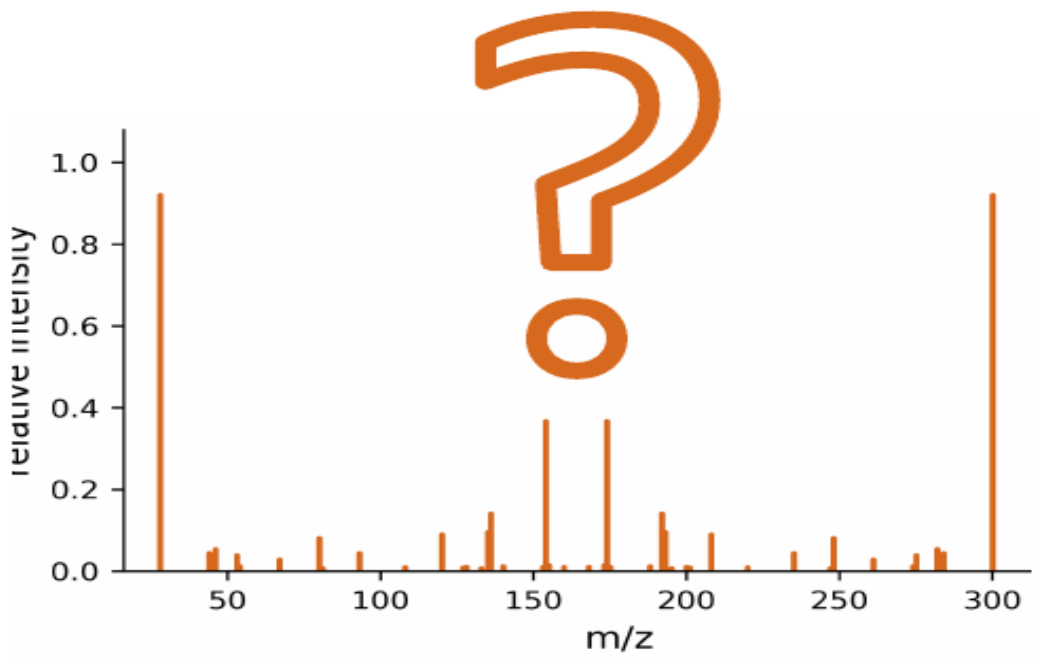
This is where the four views come together

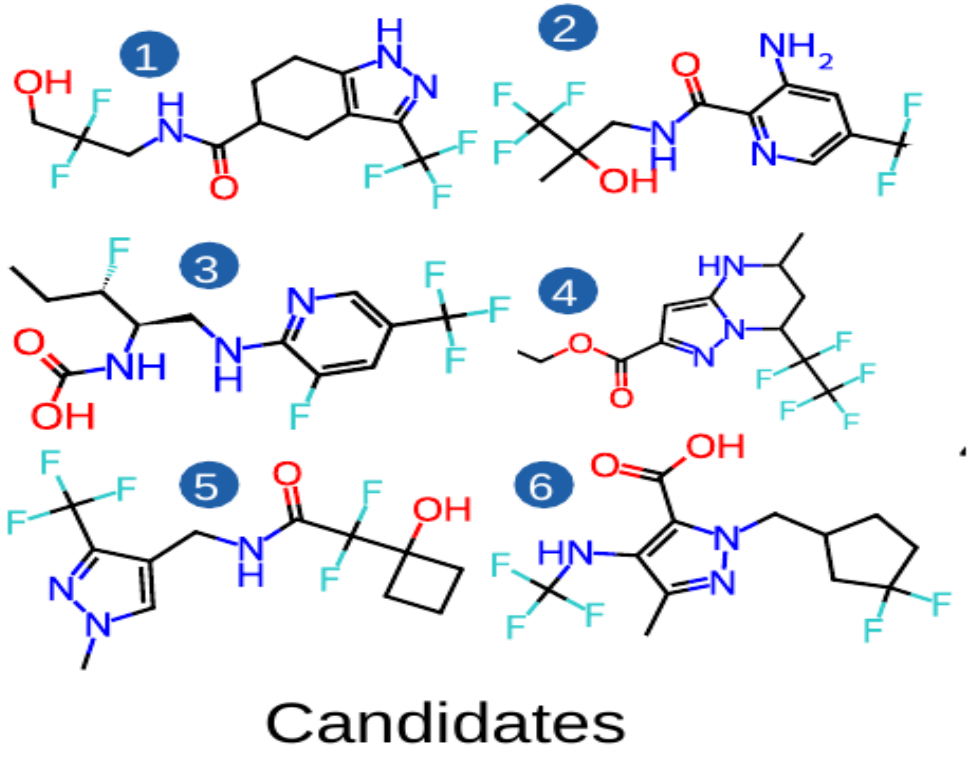
Same-compound embeddings cluster. Other compounds are pushed apart.



Each coloured cluster is one compound. The red arrow is the InfoNCE push between compounds.

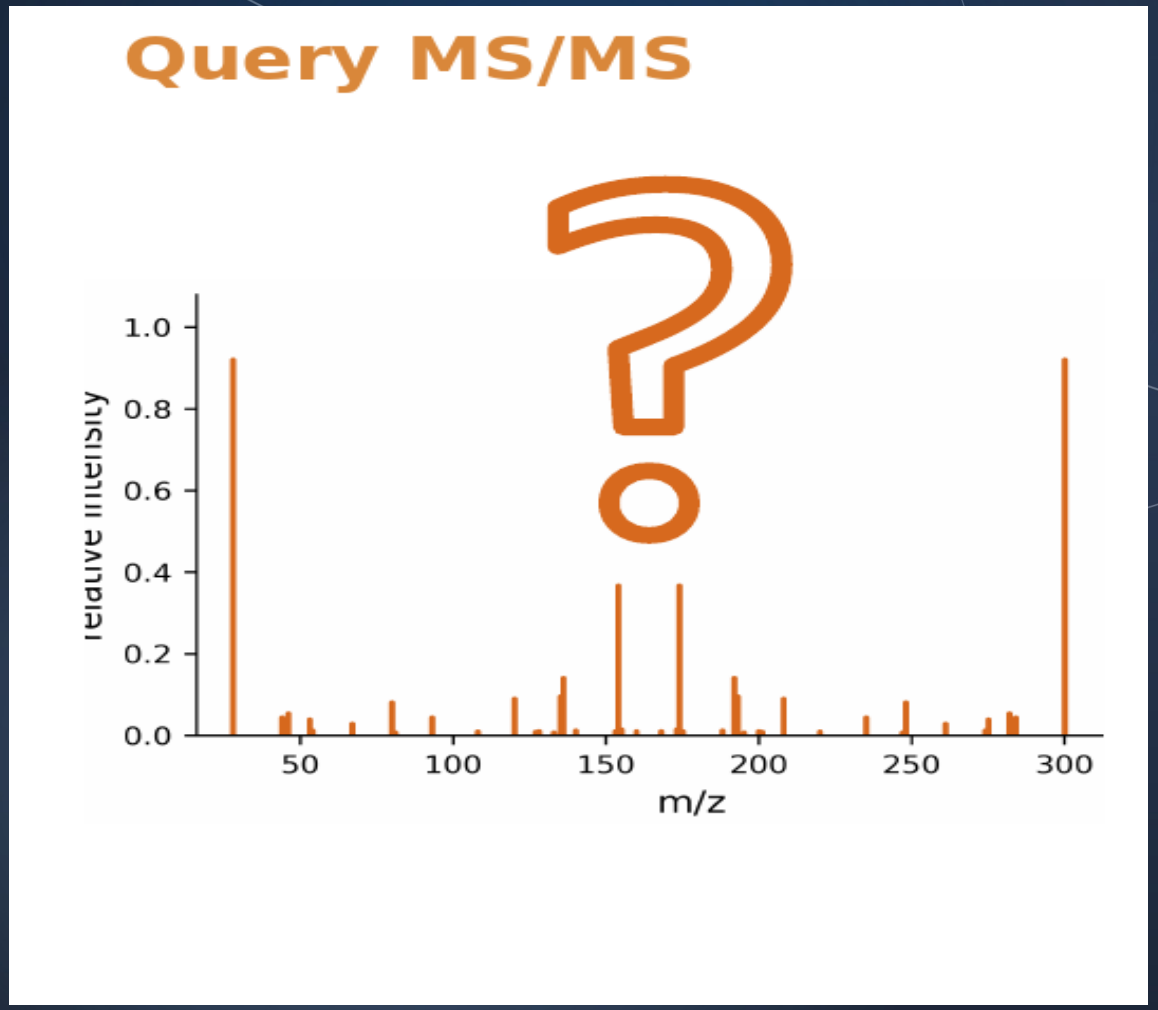
Coming back: How do we find the correct candidate?

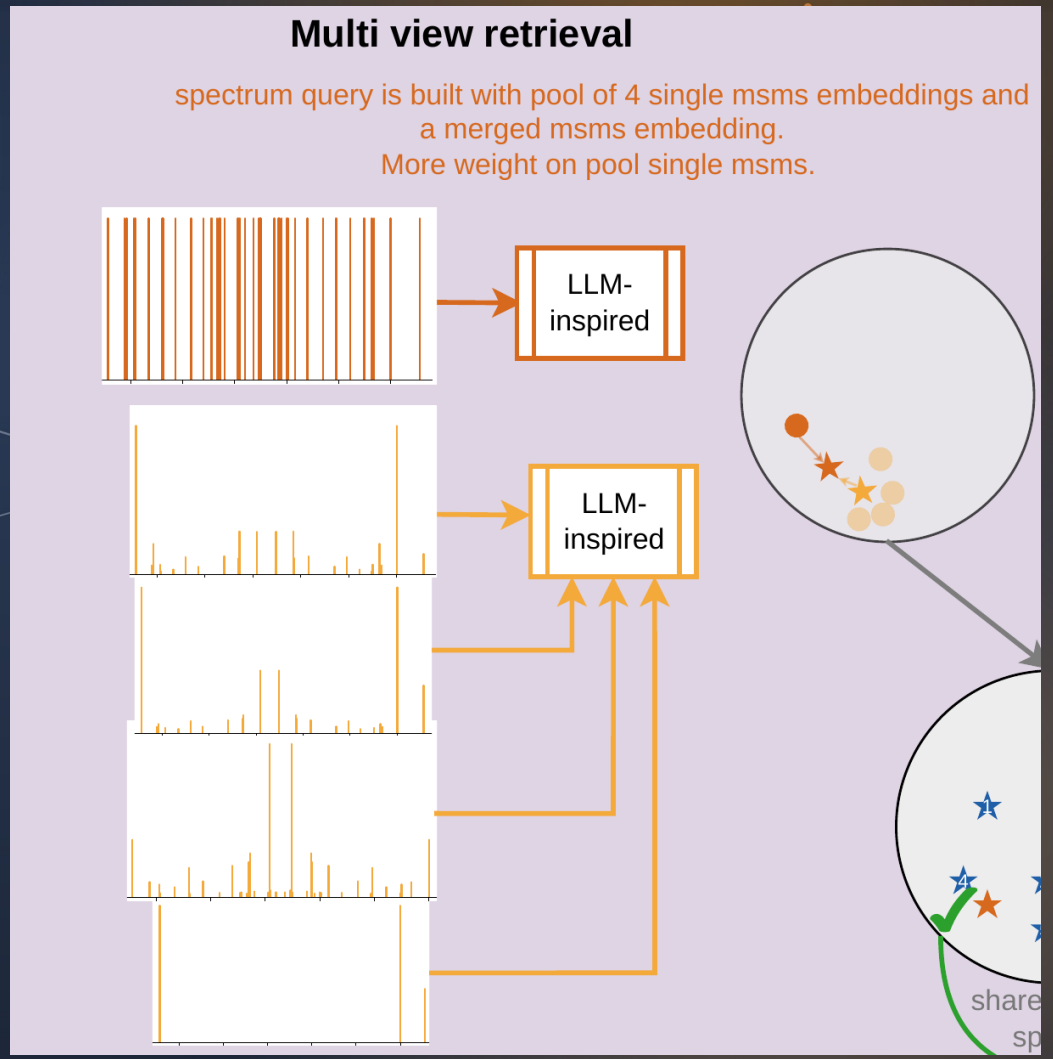




At inference: 1. we combine the spectrum embeddings

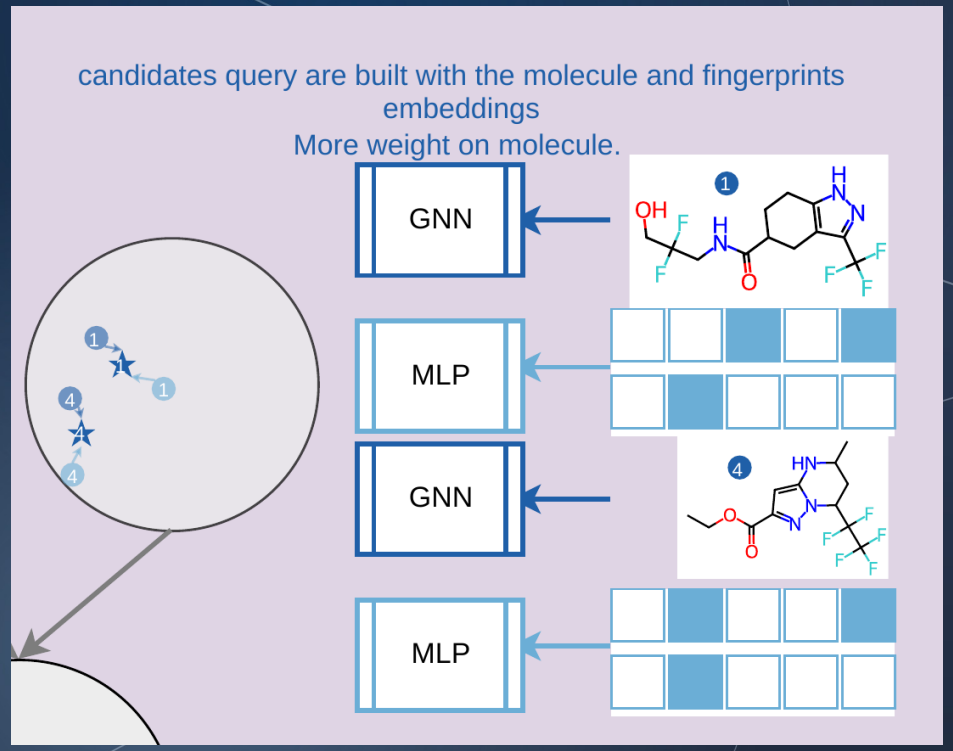
Query from merged + pooled singles

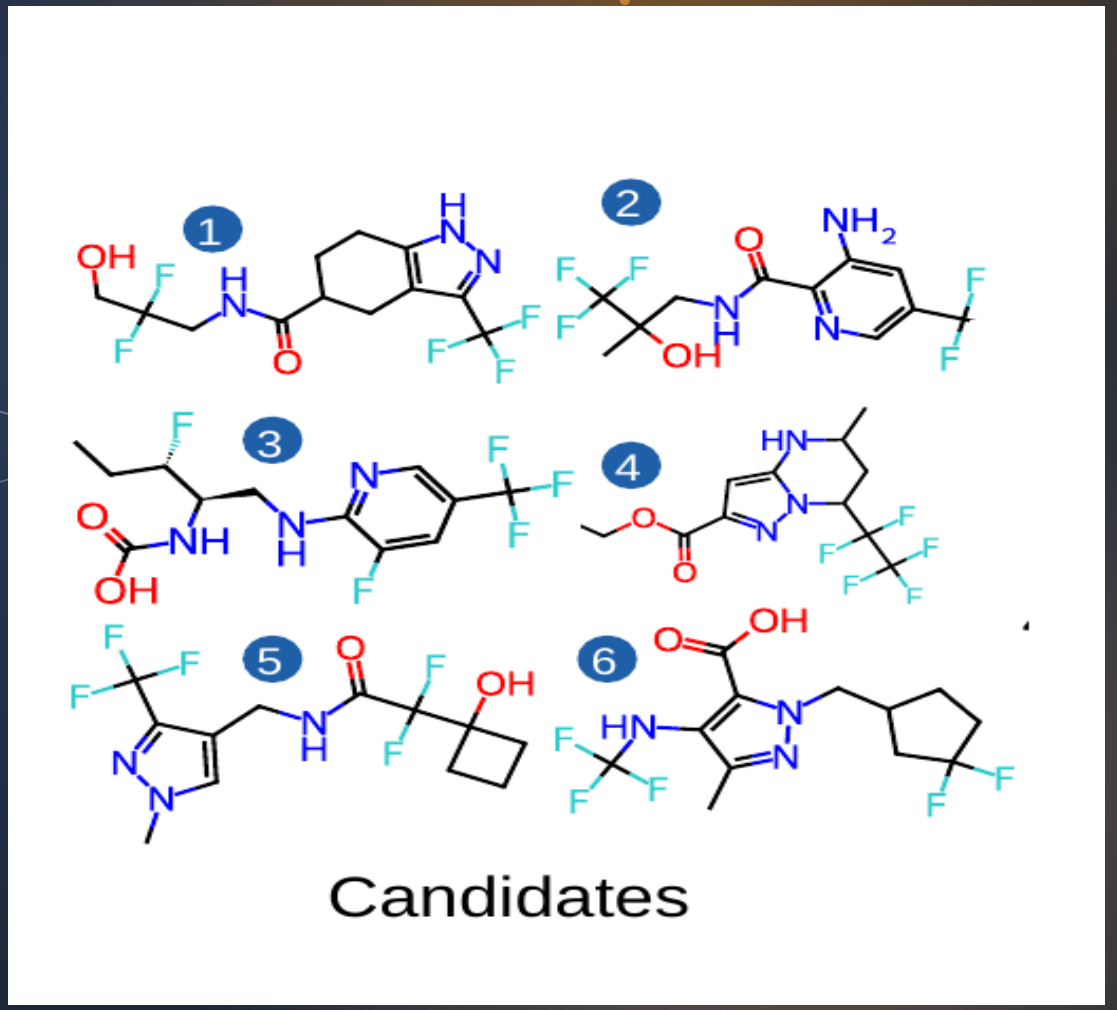




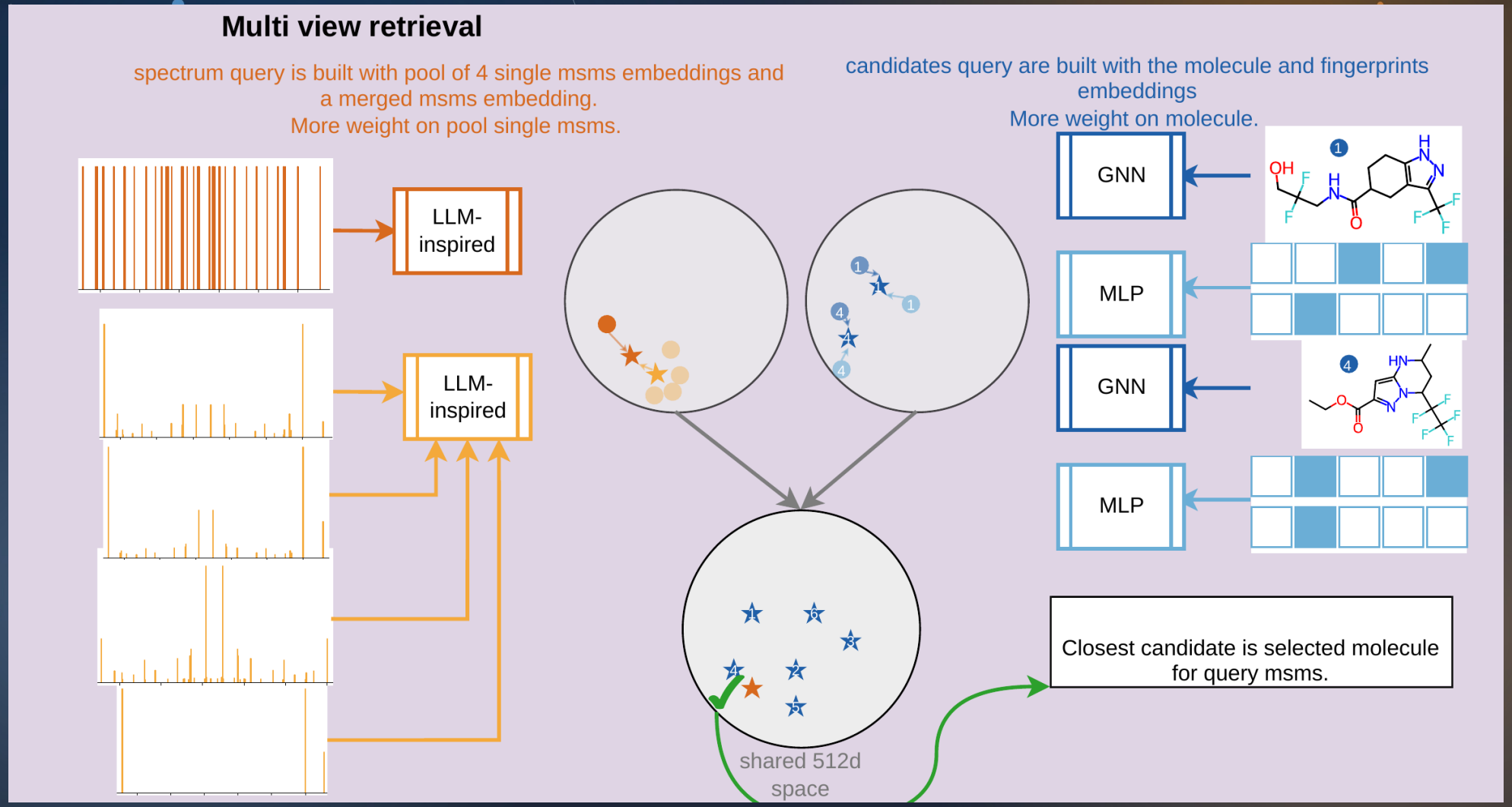
At inference: 2. we obtain the embeddings for each candidate

Each candidate from graph + fingerprint.



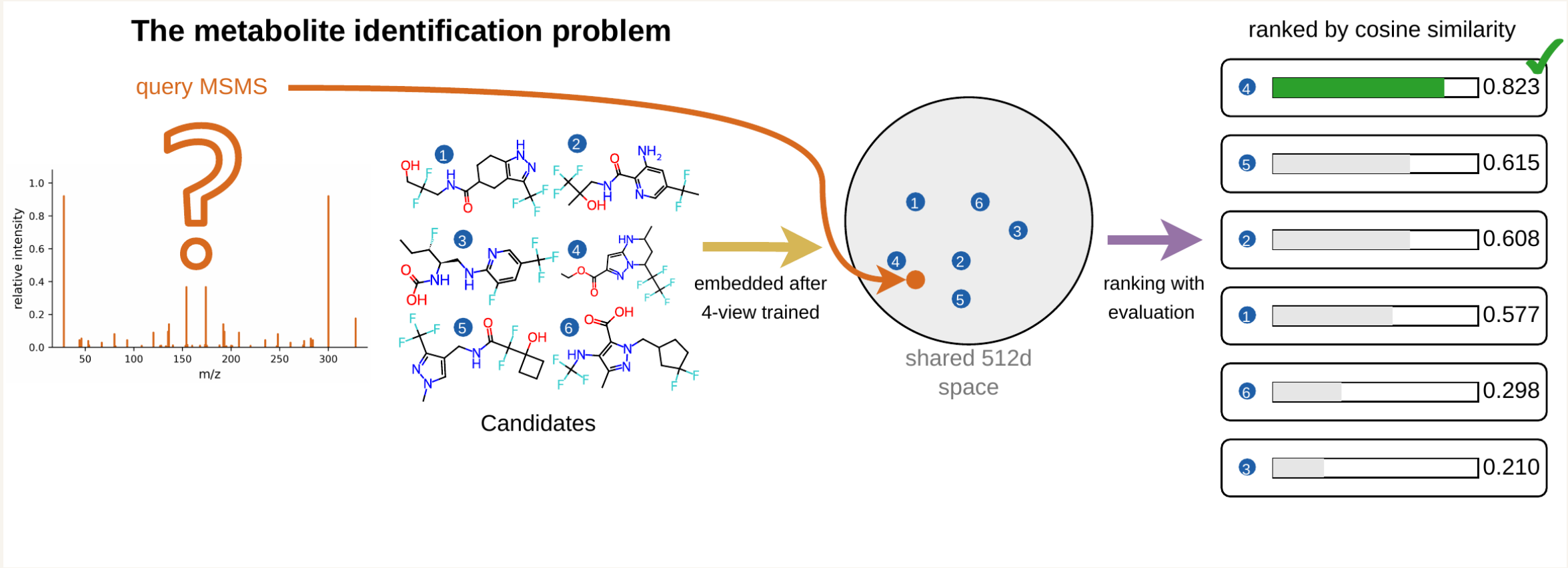


At inference: 3. we calculate the cosine distance from the query embedding to each candidate.



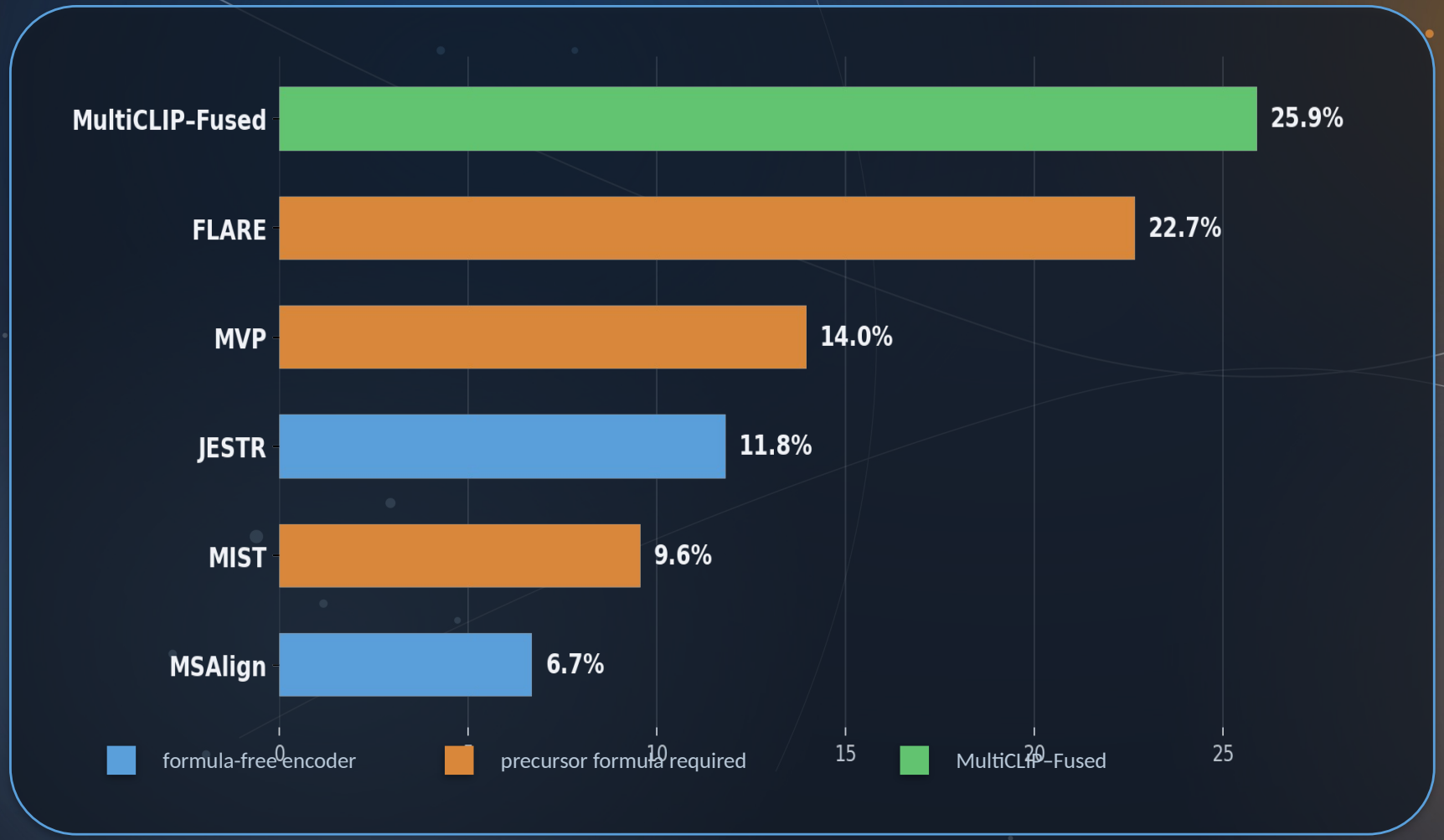
The problem, solved.

Same query · four-view embedding · the right isomer ranks first.



Best published Recall@1 — with formula-free spectrum encoding

MassSpecGym formula-known track · up to 256 isomers per query



MULTICLIP-FUSED

25.9%

Recall@1

53.8%
Recall@5

78.4%
Recall@20

Candidate list: formula-constrained
Spectrum encoder: formula-free

What does the embedding space actually look like?

01 INPUT

Tokenise m/z directly

02 ENGINE

Align and fuse four views

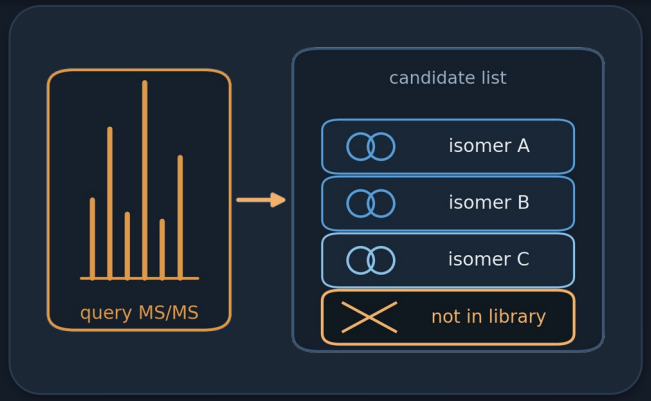
03 HORIZON

Understand the latent geometry

Motivation for understanding the geometry.

Retrieval ranks existing candidates, the future is generating new molecules with guiding embedding.

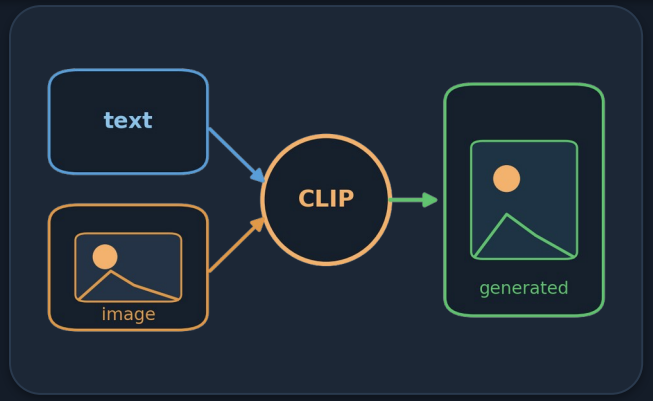
01 · DATABASE CEILING



Retrieval has a hard wall

A novel metabolite is not in the list. Ranking cannot invent it.

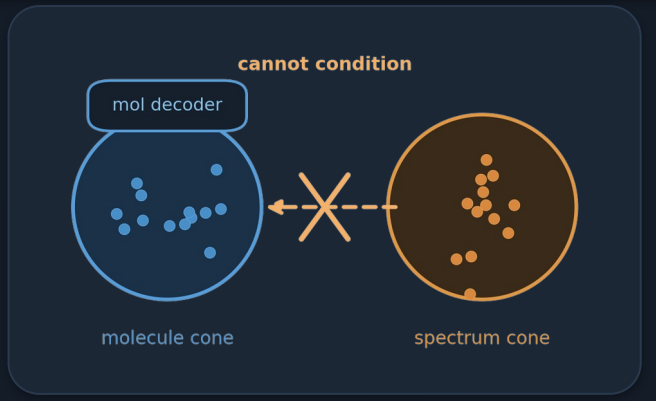
02 · OTHER CLIP DOMAINS



Vision–language already generates

unCLIP and DALL·E 2 already decode from contrastive latents.

03 · THE OBSTACLE



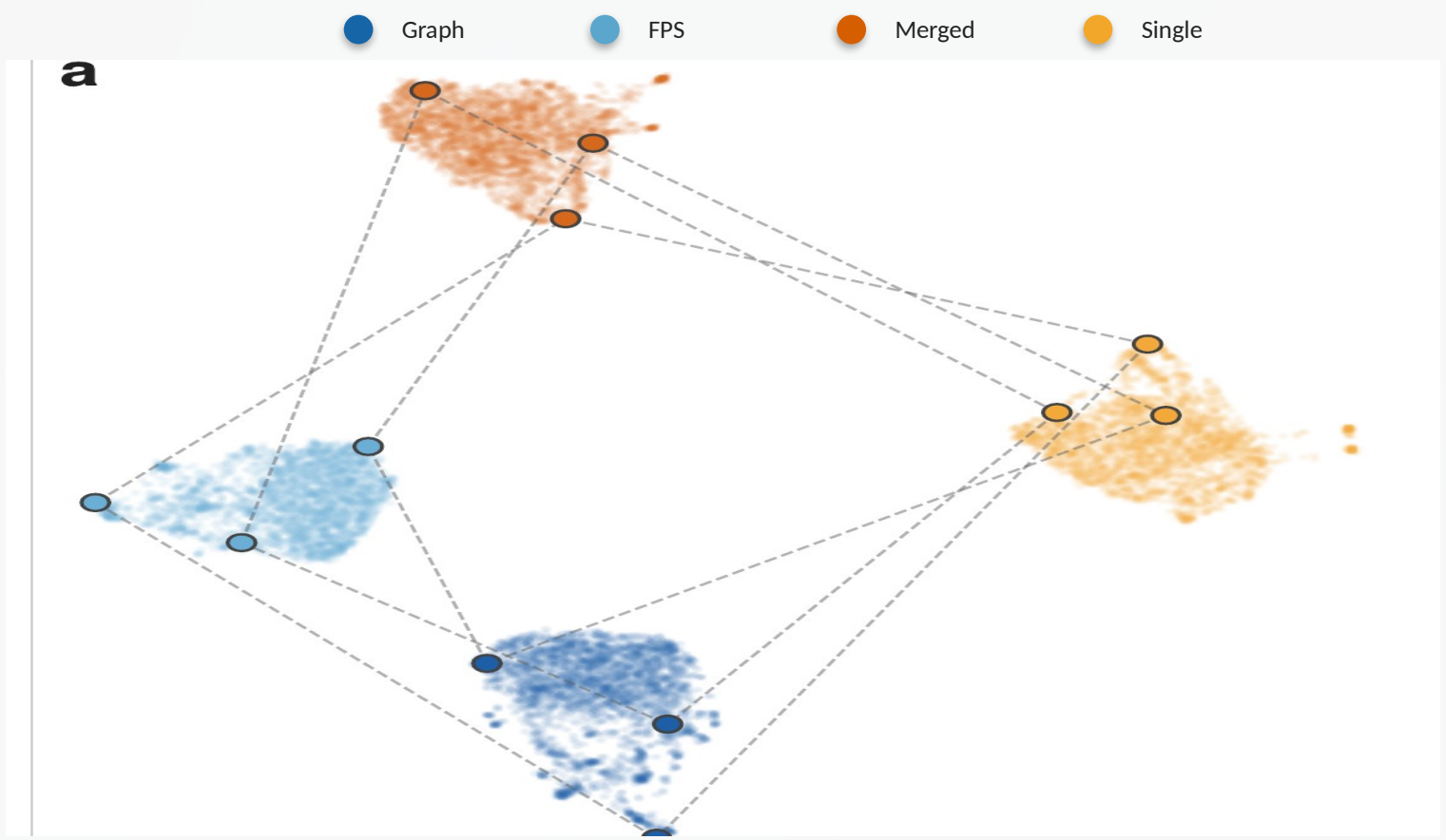
A decoder would see a shift

A graph decoder fed a spectral vector is out of distribution.

No method has yet generated structures from contrastive MS/MS embeddings. The missing piece is geometric.

Contrastive alignment does not make the modalities overlap

In the raw 512-D space, the four representations occupy separate regions.



The modality gap

Same compound.
Four embeddings.
Four separated cones.

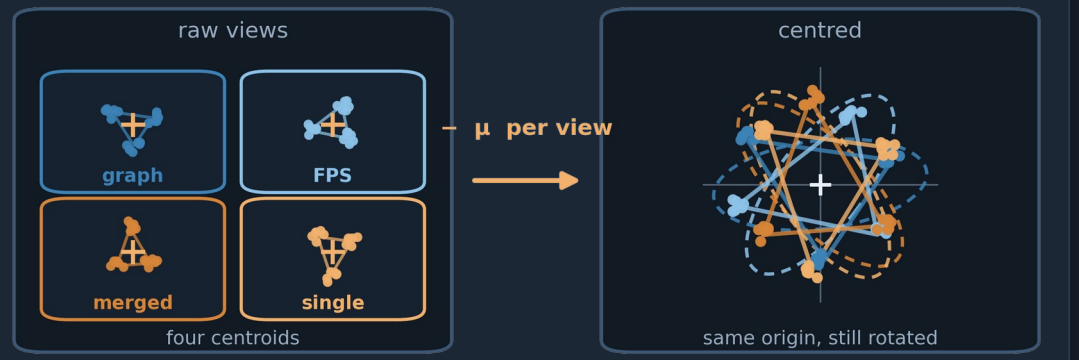
But note the pattern:

the displacement is systematic,
not arbitrary scatter.

Two deterministic steps. Nothing is retrained.

Centre each view, then rotate the others onto the graph frame.

01 · MEAN-CENTRE



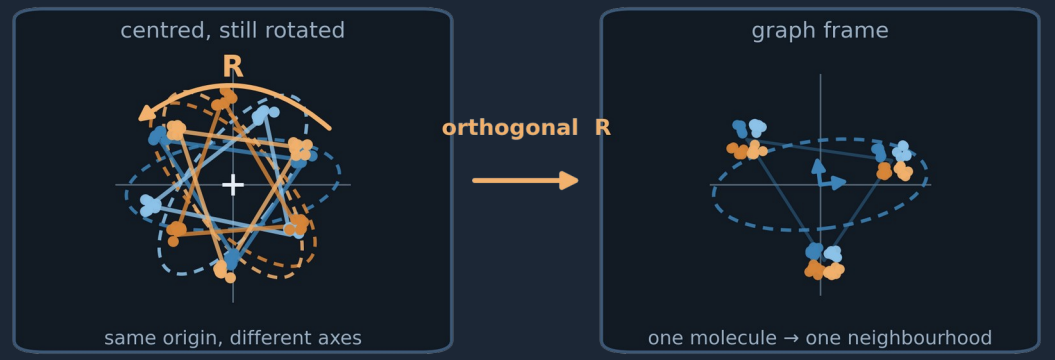
Subtract the view mean

Each encoder has its own centroid.

We move every cloud to the origin.

Relative angles inside a view are unchanged.

02 · PROCRUSTES



Rotate onto the graph frame

An orthogonal R is fitted in 512-D.

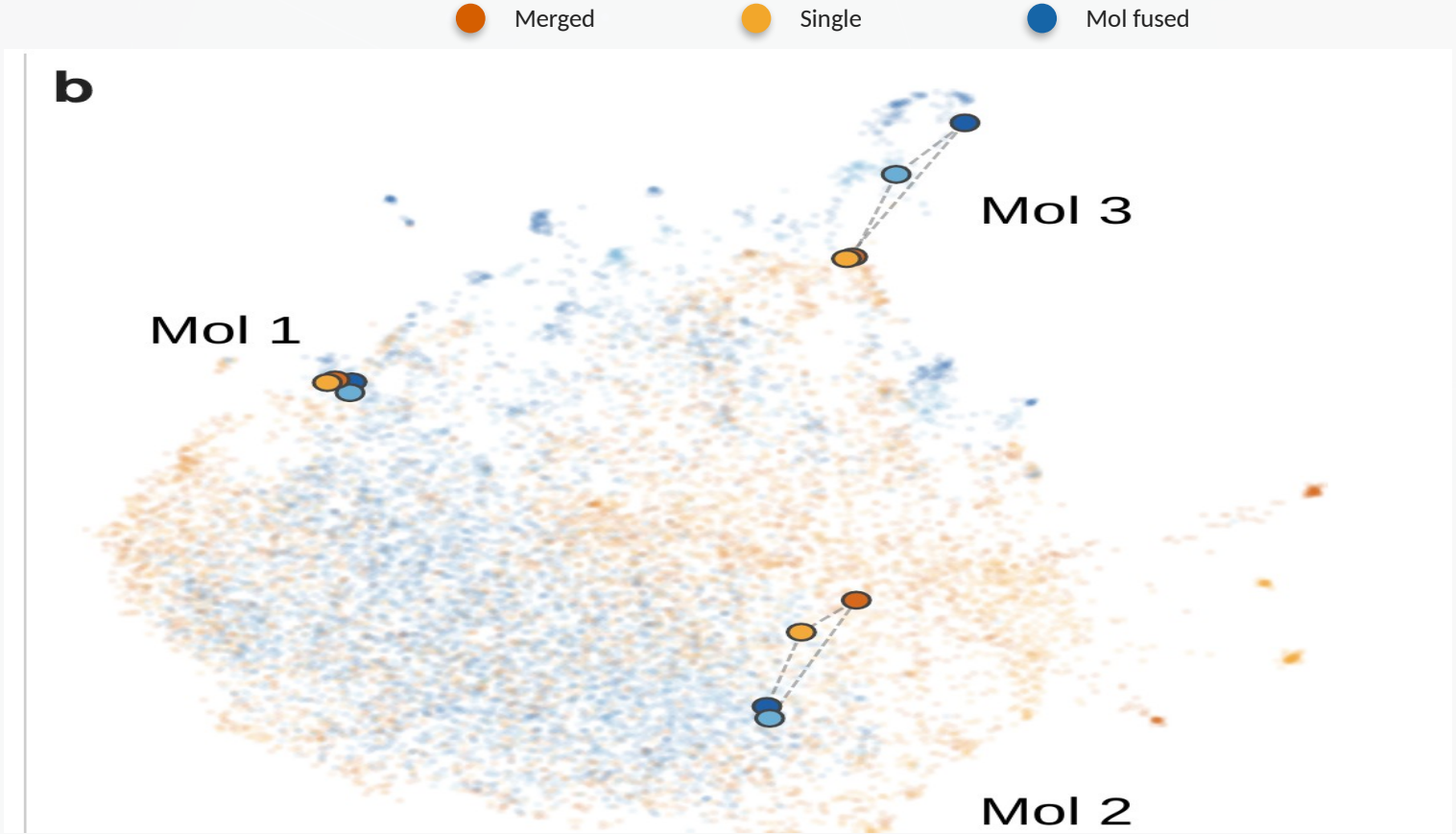
Fingerprint, merged and single rotate onto graph.

Distances inside each view are preserved.

Affine, orthogonal, no extra training. Chemistry is preserved; only the coordinate frame moves.

A deterministic alignment closes the gap

Per-view mean-centering + paired orthogonal Procrustes rotation.



No retraining

centre each view
+
rotate into one frame

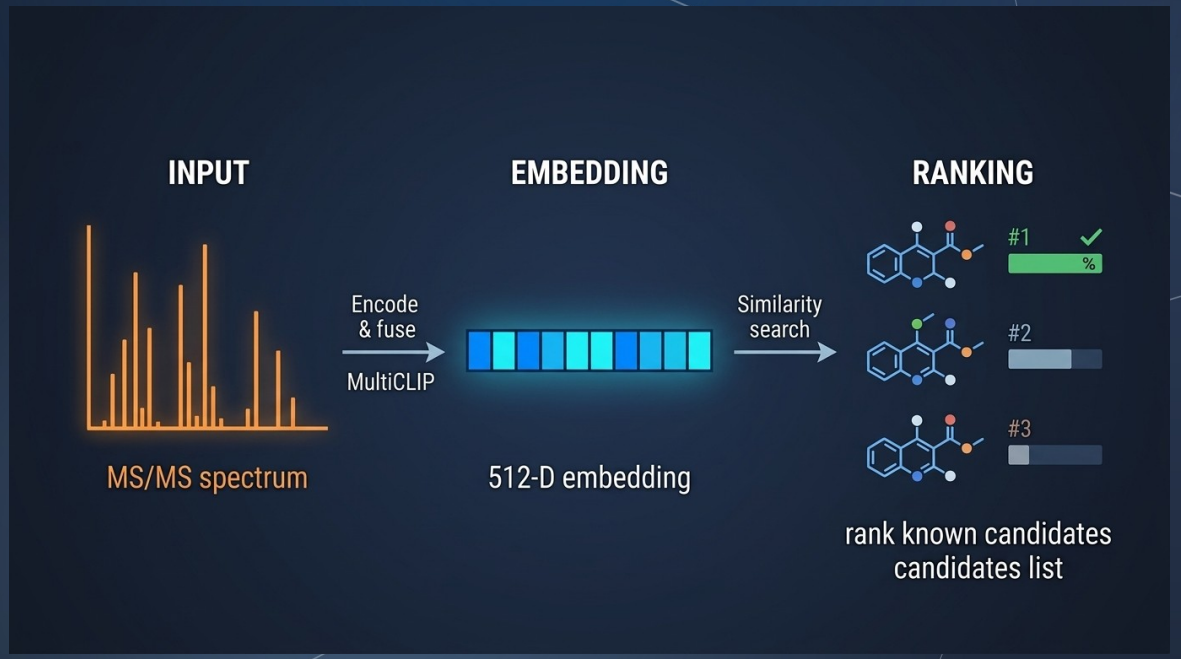
Result

the four views overlap without sacrificing structural resolution.

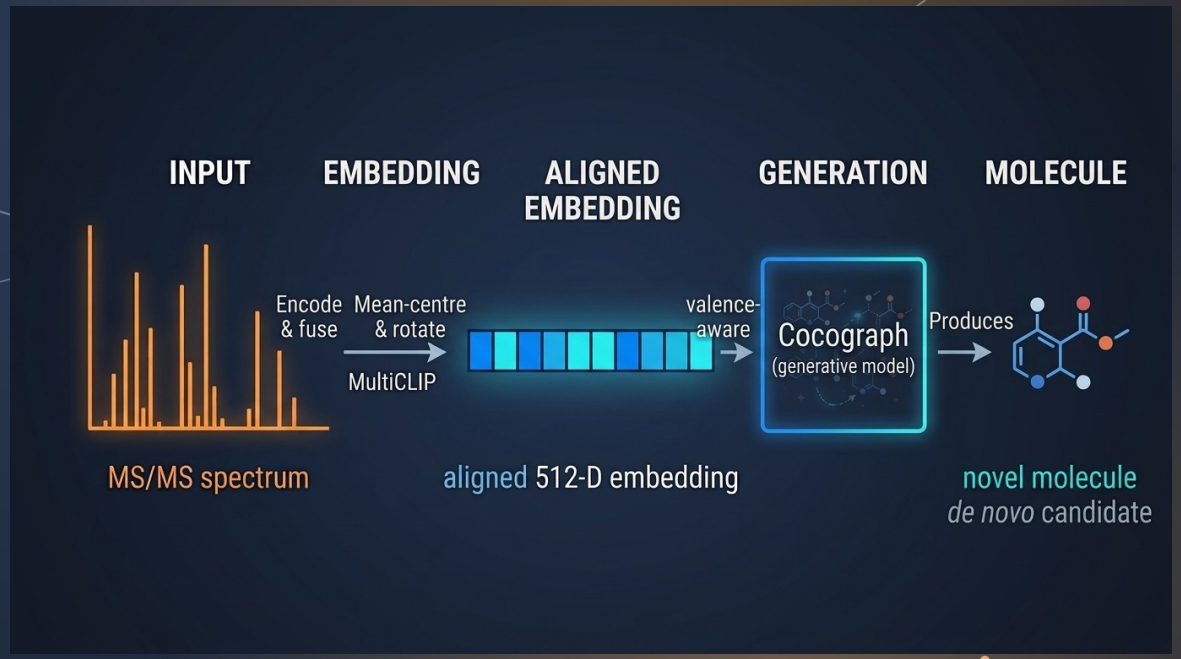
Retrieval is today's result. Generation is the next experiment.

No de novo claim: the aligned spectral embedding is a testable conditioning signal.

TODAY · RETRIEVAL



NEXT · GENERATION



The benchmark result is retrieval.
The geometric result removes a known obstacle to continuous conditioning.

We identify the isomer this way.

01 · WITHOUT THE FORMULA

m/z tokens. No precursor formula.

The peaks go in as they are measured.

02 · BY FUSING FOUR VIEWS

Graph, fingerprint, merged, single.

Align them, then fuse at inference.

03 · WITH SOTA RESULTS

25.9% Recall@1 · 53.8% @5

Best published, formula-free encoding.

THIS OPENS A NEW WAY TO RECONSTRUCT

The aligned embedding is not only a key for retrieval.
It is a continuous condition for generating molecules that are not in any library.